

Practical -01

Code Output:

```
lenovo@DESKTOP-NTSK193: ~$ nano pract1.c
lenovo@DESKTOP-NTSK193: ~$ gcc pract1.c
lenovo@DESKTOP-NTSK193: ~$ ./a.out
Enter a Linux command: cmd
Parent process
Child process
Parent process id is 511
My parent process id is 305
Child process id is 512
Parent process id is 511
Command execution failed
Child process completed
lenovo@DESKTOP-NTSK193: ~$ gcc pract1.c
lenovo@DESKTOP-NTSK193: ~$ ./a.out
Enter a Linux command: ls
Parent process
Parent process id is 539
My parent process id is 305
Child process
Child process id is 540
Parent process id is 539
FIFO1    c1.c      p1.c    p7.c      speratedirectory
a.out    c2.c      p2.c    p8.c      student1.sh.save
a1       fork1.c   p3.c    p9.c      test.txt
a1.c     fork2.c   p4.c    pract1.c  variables1.sh
a2       longerfile p5.c    round
backup  myfifo    p6.c    sender.c
Child process completed
lenovo@DESKTOP-NTSK193: ~$ |
```

b) Linux terminal commands

uname

```
lenovo@DESKTOP-NTSK193: ~$ uname
Linux
lenovo@DESKTOP-NTSK193: ~$ |
```

lscpu

```
lenovo@DESKTOP-NTSK193:~$ lscpu
Architecture:                x86_64
CPU op-mode(s):              32-bit, 64-bit
Address sizes:               39 bits physical, 48 bits virtual
Byte Order:                  Little Endian
CPU(s):                      8
On-line CPU(s) list:         0-7
Vendor ID:                   GenuineIntel
Model name:                  Intel(R) Core(TM) i7-10610U CPU @ 1.80GHz
CPU family:                  6
Model:                      142
Thread(s) per core:         2
Core(s) per socket:         4
Socket(s):                   1
Stepping:                    12
BogoMIPS:                    4608.01
Flags:                       fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht
                             syscall nx pdpe1gb rdtscp lm arch_perfmon rep_good nopl xtopology cpuid tsc_known_freq pni pclmulqdq vmx
                             ssse3 fma cx16 pdcm pcid sse4_1 sse4_2 movbe popcnt aes xsave avx f16c rdrand hypervisor lahf_lm abm 3d
                             nowprefetch ssbd ibrs ibpb stibp ibrs_enhanced tpr_shadow ept vpid ept_ad fsgsbase bmi1 avx2 smep bmi2 e
                             rms invpcid rdseed adx smap clflushopt xsaveopt xsavec xgetbv1 xsaves vnm1 md_clear flush_l1d arch_capab
                             ilities

Virtualization features:
  Virtualization:            VT-x
  Hypervisor vendor:         Microsoft
  Virtualization type:       full
Caches (sum of all):
  L1d:                       128 KiB (4 instances)
  L1i:                       128 KiB (4 instances)
  L2:                         1 MiB (4 instances)
  L3:                         8 MiB (1 instance)
NUMA:
  NUMA node(s):              1
  NUMA node0 CPU(s):         0-7
Vulnerabilities:
```

```
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                             syscall nx pdpe1gb rdtscp lm arch_perfmon rep_good nopl xtopology cpuid tsc_known_freq pni pclmulqdq vmx
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                             nowprefetch ssbd ibrs ibpb stibp ibrs_enhanced tpr_shadow ept vpid ept_ad fsgsbase bmi1 avx2 smep bmi2 e
                             rms invpcid rdseed adx smap clflushopt xsaveopt xsavec xgetbv1 xsaves vnm1 md_clear flush_l1d arch_capab
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  L3:                         8 MiB (1 instance)
NUMA:
  NUMA node(s):              1
  NUMA node0 CPU(s):         0-7
Vulnerabilities:
  Gather data sampling:      Not affected
  Ghostwrite:                Not affected
  Indirect target selection: Mitigation; Aligned branch/return thunks
  Itlb multihit:            KVM: Mitigation: Split huge pages
  L1tf:                      Not affected
  Mds:                      Not affected
  Meltdown:                 Not affected
  Mmio stale data:          Mitigation; Clear CPU buffers; SMT Host state unknown
  Old microcode:            Not affected
  Reg file data sampling:    Not affected
  Retbleed:                 Mitigation; Enhanced IBRS
  Spec rstack overflow:      Not affected
  Spec store bypass:         Mitigation; Speculative Store Bypass disabled via prctl
  Spectre v1:               Mitigation; usercopy/swapgs barriers and __user pointer sanitization
  Spectre v2:               Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSE-IBRS SW sequence; BHI SW loop, KVM SW lo
                             op
  Srbds:                    Unknown: Dependent on hypervisor status
  Tsa:                      Not affected
  Tsx async abort:          Mitigation; TSX disabled
  Vmscape:                  Not affected
```

lsblk

```
lenovo@DESKTOP-NTSK193:~$ lsblk
NAME MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda   8:0      0 356.9M  1 disk
sdb   8:16     0 159.4M  1 disk
sdc   8:32     0    2G    0 disk [SWAP]
sdd   8:48     0    1T    0 disk /mnt/wslg/distro
/
```

Ps

```
lenovo@DESKTOP-NTSK193:~$ ps
  PID TTY          TIME CMD
  305 pts/0    00:00:00 bash
  645 pts/0    00:00:00 ps
lenovo@DESKTOP-NTSK193:~$
```

top

```
lenovo@DESKTOP-NTSK193:~$ top
top - 08:13:23 up 8 min, 1 user, load average: 0.07, 0.03, 0.00
Tasks: 25 total, 1 running, 24 sleeping, 0 stopped, 0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem : 7801.6 total, 6546.4 free, 538.5 used, 868.3 buff/cache
MiB Swap: 2048.0 total, 2048.0 free,  0.0 used, 7263.1 avail Mem

  PID USER      PR  NI   VIRT   RES   SHR  S  %CPU  %MEM     TIME+ COMMAND
    1 root        20   0   24300  15292 11504 S   0.0   0.2   0:00.71 systemd
    2 root        20   0   3180   2208  2072 S   0.0   0.0   0:00.01 init-systemd(Ub
    6 root        20   0   3180   2048  1940 S   0.0   0.0   0:00.00 init
   46 root        19  -1  50420  16768 15488 S   0.0   0.2   0:00.16 systemd-journal
   78 systemd+    20   0  22416  14504 12008 S   0.0   0.2   0:00.07 systemd-resolve
   85 root        20   0  35320  12308  9272 S   0.0   0.2   0:00.43 systemd-udev
  110 root        20   0   2888   1948  1820 S   0.0   0.0   0:00.06 chronyd-starter
  115 root        20   0   4472   3140  2896 S   0.0   0.0   0:00.00 cron
  120 message+    20   0   8812   5396  4728 S   0.0   0.1   0:00.05 dbus-daemon
  147 root        20   0  44096  29768 14628 S   0.0   0.4   0:00.25 networkd-dispat
  153 root        20   0   18420  9376  8144 S   0.0   0.1   0:00.10 systemd-logind
  160 root        20   0 1866288 15468 12548 S   0.0   0.2   0:00.13 wsl-pro-service
  182 syslog      20   0 220548  5820  4516 S   0.0   0.1   0:00.08 rsyslogd
  208 _chrony      20   0  23592 11048  7092 S   0.0   0.1   0:00.07 chronyd
  221 _chrony      20   0  12096  2344  1696 S   0.0   0.0   0:00.00 chronyd
  240 root        20   0 123460 32532 17916 S   0.0   0.4   0:00.21 unattended-upgr
  251 root        20   0   5492   2720  2532 S   0.0   0.0   0:00.01 agetty
  299 root        20   0   3184   1112   980 S   0.0   0.0   0:00.00 SessionLeader
  301 root        20   0   3200   1252  1108 S   0.0   0.0   0:00.05 Relay(305)
  305 lenovo      20   0   6268   5536  3864 S   0.0   0.1   0:00.10 bash
  308 root        20   0   8644   5580  4740 S   0.0   0.1   0:00.01 login
  367 lenovo      20   0  22760 13452 10992 S   0.0   0.2   0:00.15 systemd
  369 lenovo      20   0  22924  4132  2124 S   0.0   0.1   0:00.00 (sd-pam)
  395 lenovo      20   0   6136  5420  3860 S   0.0   0.1   0:00.04 bash
  659 lenovo      20   0   8160  5984  3888 R   0.0   0.1   0:00.01 top
```